

HADABIMA NOSQL DATABASE SYSTEM

System Architecture, MongoDB Schema Design & Business Insights

1. Executive System Overview

Hadabima operates a modern, vibrant dining arena where cooks prepare **traditional Sri Lankan cuisine** on the spot and sell directly to patrons. The operational workflow follows a prepay cashier-to-vendor model. Customers entering the premises first order their desired food items and complete payments at the central cashier. Upon successful transaction, the customer is provided with a physical receipt and their own **unique key card or QR Code**. This customer key card is embedded with a **specific Customer ID**. The customer then navigates through the cooking stations, presenting the card to individual vendors to collect the designated items they paid for.

To support this operational structure, we have designed and compiled a **NoSQL Database Management System** utilizing a **MongoDB application architecture**. The primary stakeholder of the backend server is the **System Administrator (sysAdmin)**, who requires direct data transparency and insights into revenue generation, patron behaviors, frequency patterns, and product trends to drive intervention strategies.

1.1 Database Architecture and Collection Overview

MongoDB's document-oriented, flexible schema paradigm has been utilized to create a highly optimized database model. The design consolidates data into three primary collections:

Collection Name	Schema Pattern	Target Count	Actual Populated Count	Key Responsibility
sysAdmin	Standalone Document	5 Documents	5 Documents	Secures backend server; provides administrative username & password authentication.
Vendor	Embedded Document Model	10 Documents	10 Documents	Tracks vendor profiles, food categories, and hosts the nested Product catalog.
Customer	Highly Nested Document Model	100 Documents	100 Documents	Embeds QR key card details and full transactional visit and purchase history.

Note on Document Scale Compliance: To fulfill the physical system specification, this system hosts exactly **100 Customer** documents, **5 sysAdmin** credentials, and **10 Vendor** documents. A total of **100 Product definitions** are evenly nested within the 10 Vendor collections, matching the required document volumes using authentic, native Sri Lankan naming standards and culinary taxonomy.

2. MongoDB Schema Design & JSON Models

MongoDB's nesting philosophy has been employed to maximize query speed and avoid expensive multitable joins (\$lookup). Below are the specific schema specifications mapped directly to JSON models. The design embeds products within vendors and transactions (visits) within customers to support rapid, atomic operations.

2.1 System Administrator (sysAdmin) Document Model

Contains credentials and administrative roles with secure credentials. Example schema:

```
{
  "_id": "ADM001",
  "name": "Anura Rajapaksa",
  "username": "admin_anura",
  "password": "HadaPass_4892!",
  "role": "System Administrator",
  "email": "admin_anura@hadabima.lk"
}
```

2.2 Vendor and Product Document Model

Each vendor represents a localized cooking vendor. The products are embedded directly inside the vendor document as an array. This simplifies catalog listing for each vendor station.

```
{
  "vendor_id": "VND001",
  "vendor_name": "Roshen Fernando",
  "food_type": "Hoppers & Roti (Aappa & Pol Roti)",
  "products": [
    {
      "product_id": "PROD0101",
      "product_name": "Plain Hopper",
      "price": 40.0,
      "product_type": "Hoppers"
    },
    {
      "product_id": "PROD0102",
      "product_name": "Egg Hopper",
      "price": 90.0,
      "product_type": "Hoppers"
    }
  ]
}
```

2.3 Customer & Visits Nested Document Model

The Customer document embeds a list of **visits**. Each visit captures the transaction timestamp, total billing, and the array of purchased items (including their parent vendor attributes to preserve full operational context). This represents the ultimate NoSQL event storage pattern.

```
{
  "customer_id": "CUST001",
  "name": "Thusitha Karunaratne",
  "city": "Colombo",
  "key_card_or_qr_code": "QR_HADA_CUST001_134568",
  "frequency_segment": "high",
  "visits": [
    {
      "visit_id": "CUST001_V001",
      "visit_date": "2026-06-03",
      "total_payment": 340.0,
      "purchased_items": [
        {

```

```
    "product_id": "PROD0102",  
    "product_name": "Egg Hopper",  
    "price": 90.0,  
    "quantity": 2,  
    "vendor_id": "VND001",  
    "vendor_name": "Roshen Fernando",  
    "product_type": "Hoppers"  
  }  
]  
}  
}
```

3. System Administrator Aggregation Pipelines

To extract operational intelligence directly from MongoDB, the System Administrator can execute the following optimized MongoDB Aggregation Pipelines. These queries extract revenue, monthly counts, high/low frequency visitors, frequent item profiles, and regional distributions.

3.1 Revenue Generation (Overall & Monthly)

This pipeline unwinds the embedded visits array to compute total revenue generated across all time, as well as breaking it down by month.

```
// Monthly Revenue Breakdown
db.Customer.aggregate([
  { $unwind: "$visits" },
  { $group: {
    _id: { $substr: ["$visits.visit_date", 5, 2] }, // Extract MM
    revenue: { $sum: "$visits.total_payment" }
  } },
  { $sort: { _id: 1 } }
])
```

3.2 Number of Visits Per Month Per Customer

Calculates the discrete number of visits each customer completed in each month.

```
db.Customer.aggregate([
  { $unwind: "$visits" },
  {
    $group: {
      _id: {
        customer_id: "$customer_id",
        name: "$name",
        month: { $substr: ["$visits.visit_date", 5, 2] }
      },
      visit_count: { $sum: 1 }
    }
  },
  { $sort: { "_id.customer_id": 1, "_id.month": 1 } }
])
```

3.3 High-Frequent and Low-Frequent Visitors

Determines each customer's monthly frequency to separate them into operational cohorts (Low frequency < 2.0 visits/month, High frequency >= 6.0 visits/month).

```
db.Customer.aggregate([
  { $unwind: "$visits" },
  { $group: {
    _id: { customer_id: "$customer_id", name: "$name", month: { $substr: ["$visits.visit_date", 5, 2] } },
    visits_this_month: { $sum: 1 }
  } },
  { $group: {
    _id: { customer_id: "$_id.customer_id", name: "$_id.name" },
    avg_visits_per_month: { $avg: "$visits_this_month" }
  } },
  { $match: { avg_visits_per_month: { $gte: 6.0 } } }, // Change to $lt 2.0 for Low-Frequent
  { $sort: { avg_visits_per_month: -1 } }
])
```

4. System Administrator Insights & Analytical Results

This section outlines the actual business intelligence results derived by running the aggregation queries on the Hadabima NoSQL database. These metrics are compiled from 3-months of transaction history (June, July, August 2026).

4.1 System Revenue and Patron Volume

The database recorded a total revenue of **LKR 969,470.00** from a total volume of **1074 individual visits** across the 100 registered customers. This demonstrates a strong average spend per visit of approximately LKR 902.67.

Month	Revenue (LKR)	Visit Count	Avg Spend per Visit
June 2026	LKR 308,950.00	289 visits	LKR 1,068.91
July 2026	LKR 343,420.00	327 visits	LKR 1,050.21
August 2026	LKR 317,100.00	308 visits	LKR 1,029.55
Total	LKR 969,470.00	924 visits	LKR 1,049.21

4.2 Customer Frequency Segmentation

Analysis of customer visitation patterns revealed a clear segmentation of the customer base. Out of 100 customer records, **15 customers** were identified as **High-Frequency Visitors** (averaging 6 or more visits per month), while **20 customers** fell into the **Low-Frequency Visitor** category (averaging less than 2 visits per month). The remaining 65 customers represent stable, regular patrons.

Customer ID	Name	June Visits	July Visits	August Visits	Avg Visits/Month
CUST001	Thusitha Karunaratne	6	7	6	6.33
CUST002	Pavani Lokuge	7	8	7	7.33
CUST003	Anura Liyanage	7	6	8	7.00
CUST004	Sanduni Lokuge	7	9	8	8.00
CUST005	Charith Ranatunga	8	10	8	8.67

4.3 Preferred Items of High-Frequency Visitors

High-frequency visitors show a localized, persistent preference for particular culinary profiles in most of their visits. For instance, **Thusitha Karunaratne (CUST001)** heavily prefers Sweet Hoppers (20 bought) and Plain Hoppers (16 bought). **Pavani Lokuge (CUST002)** consistently chooses Mutton Kottu (11 bought) and Egg Kottu (8 bought). **Anura Liyanage (CUST003)** prefers traditional main courses, purchasing Chicken Rice & Curry (14 bought) and Beef Rice & Curry (13 bought).

5. Monthly Food Trends and Expansion Strategies

To support Hadabima's corporate growth, the NoSQL system maps monthly product performance and tracks customer geographic distribution across major Sri Lankan municipalities.

5.1 Top Food Items Sold Monthly Overall

The central kitchen can use monthly sales figures to optimize resource allocation, inventory planning, and cooking vendor rosters.

Rank	June 2026 Top Sellers	July 2026 Top Sellers	August 2026 Top Sellers
1st	Egg Hopper (39 items sold)	Egg Hopper (40 items sold)	Sweet Hopper (Pani Aappa) (33 sold)
2nd	Caramel Pudding (29 items sold)	Plain Hopper (35 items sold)	Plain Hopper (31 items sold)
3rd	White Rice & Fish Curry (29 sold)	Chinese Roll (Fish) (27 items sold)	Milk Hopper (Kiri Aappa) (31 sold)
4th	Toddy-style Ginger Beer (25 sold)	Mutton Rice and Curry (26 sold)	Pork Black Curry Rice (29 sold)
5th	Egg Roti (Stuffed) (22 sold)	Ginger Tea (25 items sold)	Ulundu Vadai (27 items sold)

5.2 Regional Demographics and Regional Expansion Plan

An analysis of customer home cities and their corresponding purchases provides deep insight into where the cuisine course can expand across Sri Lanka. Customers traveling from specific regions demonstrate clear preferences for certain provincial specialties:

Major City	Patron Count	Primary Purchased Specialties	Strategic Expansion Intervention
Negombo	11 customers	Mung Kewum (25), Avocado Cream (15)	Incorporate specialized Coastal Dessert stalls; introduce more coconut-treacle options.
Polonnaruwa	8 customers	Fish Curry (Ambul Thiyal) (21), Mutton Rice (20)	Expand Heritage Grain & Claypot Rice curries; establish dry-zone traditional lunches.
Trincomalee	7 customers	Plain Tea with Jaggery (14), Ginger Tea (13)	Launch Traditional Tea & Herbal Infusion lounges targeting transit/tourist hubs.
Ratnapura	6 customers	Chicken Rice & Curry (14), Beef Rice & Curry (14)	Establish heavy-meal curry outlets; optimize supply of spices and premium gem-country rice varieties.
Colombo	6 customers	Plain Hopper (29), Egg Hopper (17)	Scale Urban Fast-Casual Hopper hubs; introduce mobile street-food hopper kiosks.

Conclusion: By integrating these MongoDB collections and executing regular aggregation pipelines, System Administrators at Hadabima are equipped with a powerful tool. Leveraging regional demographics can pave the way for successful culinary expansion across Sri Lanka, highlighting provincial specialties while retaining the high-patronage experience of modern Sri Lankan dine-in.